
Cost-regularised ansatz growth: adaptive variational quantum dynamics native to a fixed qubit topology

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Abstract

Adaptive variational quantum dynamics simulation (AVQDS) builds its circuit as it goes: whenever McLachlan’s variational principle can no longer track the exact evolution, a new Pauli rotation is appended, chosen from an operator pool to minimise the McLachlan distance. The resulting circuits are one to two orders of magnitude shallower than Trotterisation, but they assume all-to-all connectivity, and the two-qubit rotations they request are generally not executable on a real device. We show that the freedom hidden inside the growth step is enough to remove most of that mismatch. At every growth event AVQDS already computes the post-insertion McLachlan distance for every pool operator; typically many operators are *acceptable* — either individually sufficient to bring L^2 below threshold, or within a small tolerance band of the best — and the algorithm’s choice among them is arbitrary from the standpoint of accuracy. We make that choice on hardware grounds instead, selecting the acceptable operator that is cheapest to place under the current logical-to-physical qubit assignment, with SWAPs inserted online as the ansatz grows. The resulting method, cost-regularised ansatz growth (CRAG-AVQDS), is a compact acceptance rule layered on an otherwise unmodified AVQDS core, with a single hyperparameter and a switch that recovers the original algorithm bit-for-bit. On the eight-site mixed-field Ising and Lieb–Schultz–Mattis benchmarks, and on three topologies — a 2×4 square lattice, an eight-qubit heavy-hex subgraph and a linear chain — CRAG-AVQDS is fully topology-compliant, matches unconstrained AVQDS fidelity to 0.030 % absolute in the worst case, and cuts the CNOT cost of connectivity on the heavy-hex tree, where no Hamiltonian path exists and routing is unavoidable, from $1.87\times$ for post-hoc SWAP routing — or $1.81\times$ when that baseline is handed the qubit placement that minimises its own realised cost, found by exhaustive search over all $8!$ assignments — down to $1.24\times$. The selection rule itself is free (0.001% of wall time), though the larger ansatz it builds costs $1.25\times$ the total classical time. Ablation is candid about where the gain comes from: the ansatz both uses fewer two-qubit rotations and places them more locally, and it is the first, topology-*independent* effect that dominates — graph-distance awareness contributes only 2% of the margin over the oracle-placed baseline, and on 2 of the 4 ablated benchmarks it is net negative. The connectivity-specific gain is real but smaller than the headline multiplier alone implies.

1 Introduction

Simulating the real-time evolution $|\Psi(t)\rangle = e^{-i\hat{H}t} |\Psi(0)\rangle$ of a quantum many-body system is among the few tasks for which quantum hardware is expected to offer a decisive advantage Feynman [1982], Lloyd [1996]. On pre-fault-tolerant devices the obstacle is circuit size: a first-order Trotter circuit grows linearly with the simulated time and quickly exceeds the coherence budget Childs et al. [2021], Preskill [2018]. Variational quantum algorithms Cerezo et al. [2021], Bharti et al. [2022] replace the

product formula with a parametrised state whose parameters follow an equation of motion derived from a time-dependent variational principle [McLachlan \[1964\]](#), [Li and Benjamin \[2017\]](#), [Yuan et al. \[2019\]](#), [Barison et al. \[2021\]](#), an approach that also covers imaginary time [McArdle et al. \[2019\]](#) and general open-system processes [Endo et al. \[2020\]](#). The difficulty is then transferred to the ansatz: a fixed circuit form that is expressive enough at $t = 0$ generally is not at later times, because the state migrates into regions of Hilbert space the ansatz cannot reach. Hardware-efficient ansätze [Kandala et al. \[2017\]](#) sidestep the compilation problem by building only from native entanglers, at the cost of expressibility and of trainability in the barren-plateau sense [McClean et al. \[2018\]](#); fixing the circuit form in advance is precisely what adaptive methods avoid.

Adaptive variational quantum dynamics simulation (AVQDS) [Yao et al. \[2021\]](#) resolves this by growing the circuit along the trajectory. The state is a pseudo-Trotter product

$$|\Psi[\theta]\rangle = \prod_{\mu=0}^{N_\theta-1} e^{-\frac{i}{2}\theta_\mu \hat{A}_\mu} |\Psi_0\rangle, \quad (1)$$

of Pauli rotations drawn from an operator pool $\{\hat{A}_\mu\}$; whenever the McLachlan distance L^2 — the residual of the variational equations of motion — exceeds a threshold L_{cut}^2 , the pool is scanned, the operator whose insertion reduces L^2 the most is appended with $\theta = 0$ (which leaves the state unchanged but enlarges the tangent space), and the scan repeats until $L^2 < L_{\text{cut}}^2$. The method reaches fidelities above 99.5% on eight-site spin chains with two orders of magnitude fewer CNOTs than Trotterisation at matched accuracy [Yao et al. \[2021\]](#), and the same adaptive idea has since been carried to imaginary time, finite temperature and response functions [Gomes et al. \[2021\]](#), [Getelina et al. \[2023\]](#), [Mootz et al. \[2024\]](#), and independently to projected variational dynamics [Linteau et al. \[2024\]](#). Related non-variational routes to the same physics include quantum imaginary-time evolution [Motta et al. \[2020\]](#) and adaptive product formulas [Zhang et al. \[2023\]](#), the latter sharing our observation that the ordering and choice of exponentials is a free resource.

Those CNOT counts, however, are *logical*. AVQDS’s pool contains two-qubit Pauli strings on whatever pairs of sites appear in the Hamiltonian, and the algorithm is free to place a rotation on any pair at any time. Real processors are sparse graphs: IBM’s superconducting devices use a heavy-hex lattice of degree at most three [Chamberland et al. \[2020\]](#), [Kim et al. \[2023\]](#), and square-lattice devices are degree four. A two-qubit gate between non-adjacent qubits must be realised by moving the logical information with SWAPs, each costing three CNOTs, and the resulting overhead is the central practical tax on near-term algorithms. A large compilation literature attacks this problem — heuristic routers such as SABRE [Li et al. \[2019\]](#), the routing passes of T|KET [Cowtan et al. \[2019\]](#), [Sivarajah et al. \[2021\]](#), exact mappers [Wille et al. \[2019\]](#), [Zulehner et al. \[2019\]](#), architecture-aware CNOT and phase-polynomial synthesis [Nash et al. \[2020\]](#), [Kissinger and Meijer-van de Griend \[2020\]](#), [Cowtan et al. \[2020a\]](#), and noise-aware allocation [Murali et al. \[2019\]](#), [Tannu and Qureshi \[2019\]](#) — but all of it shares one premise: *the circuit is given, and the compiler must transport it onto the device*.

An adaptive algorithm breaks that premise. In AVQDS the circuit is not given; it is being written, gate by gate, by a procedure that at each step has a *set* of acceptable continuations rather than a unique one. Two facts make this set large. First, the growth loop’s own stopping criterion is an inequality: any operator that pushes L^2 below L_{cut}^2 discharges the algorithm’s accuracy obligation, and among such operators AVQDS’s choice is already governed by a secondary heuristic (the original implementation prefers the best-conditioned candidate). Second, as Yao *et al.* themselves observed, “the list of new McLachlan distances $\{L_\nu^2\}$ often contains almost degenerate minimal values among several operators” [Yao et al. \[2021\]](#). The accuracy criterion, in other words, does not pin down the circuit. Something else may as well.

Contribution. We use the device’s connectivity graph as that something else. The method, *cost-regularised ansatz growth* (Crag-AVQDS), replaces the acceptance rule inside AVQDS’s growth loop by a lexicographic rule that first restricts to operators acceptable to McLachlan’s principle and then, among those, selects the one with the smallest routing cost under the current logical-to-physical qubit assignment; SWAPs are inserted online, so the cost oracle always reflects the placement the circuit has actually reached. Everything else — the equations of motion, the Tikhonov regularisation, the adaptive time step, the forward-Euler update, the canonical Hamiltonian operator pools — is the unmodified AVQDS core of Ref. [Yao et al. \[2021\]](#), whose reference implementation we import verbatim. The topology layer has one hyperparameter, a tolerance band η , and setting $\eta = 0$ together

with an all-to-all graph reproduces the original algorithm bit-for-bit (verified: identical step count, identical operator sequence, $\max |\Delta\theta| = 0$).

We report three findings.

1. **Connectivity is nearly free when the interaction graph embeds.** On a 2×4 square lattice, which admits a Hamiltonian cycle, both the periodic Ising ring and the open XY chain embed with every pool bond on a device edge. Post-hoc routing is then trivially exact and both methods sit at $1.00\times$ CNOT overhead — except that CRAG-AVQDS goes *below* it, to $0.66\times$, by preferring single-qubit generators whenever they are acceptable. This gain is topology-independent and we isolate it by ablation.
2. **Where routing is unavoidable, choosing the gate beats routing it.** The heavy-hex subgraph is a tree with four leaves, so no Hamiltonian path exists and, by exhaustive enumeration over all $8!$ placements, at least two bonds of the open XY chain and at least three of the periodic Ising ring are non-local under *every* placement. Here post-hoc SWAP routing of the unconstrained circuit costs $1.87\times$ the CNOTs on average ($1.81\times$ even when handed the placement that minimises its own realised cost, found by exhaustive search over all $8!$ assignments), whereas CRAG-AVQDS costs $1.24\times$ — a 32% reduction against the stronger baseline. The mechanism is measurable: the fraction of two-qubit rotations that are already nearest-neighbour *at the moment they are emitted* rises from 57% to 68%. Ablation splits the advantage into 98% from preferring weight-one generators — a saving that is real but has nothing to do with connectivity — and 2% from graph-distance awareness; we had expected the latter to dominate on a tree, and report that it does not — on one benchmark the distance term is net negative. The connectivity-specific part of the margin is therefore considerably smaller than the headline multiplier alone suggests.
3. **The accuracy is not paid for.** Across all 12 (benchmark, topology) cells the final-time fidelity of CRAG-AVQDS differs from the unconstrained baseline by at most 0.030% absolute — far inside the 1% tolerance, and higher rather than lower in 5 of them — because the growth loop still enforces $L^2 < L_{\text{cut}}^2$ irrespective of *which* acceptable operator is used. The topology layer itself is free — a table lookup per candidate, 0.001% of wall time — though the larger ansatz it produces costs $1.25\times$ the total classical time on average (0.81–1.91 across benchmarks), since an AVQDS step scales as N_θ^2 .

Relation to prior work. Routers such as SABRE [Li et al. \[2019\]](#), the T|KET passes [Cowtan et al. \[2019\]](#), [Sivarajah et al. \[2021\]](#) and exact mappers [Wille et al. \[2019\]](#), [Siraichi et al. \[2018\]](#) choose SWAPs for a fixed gate list; compilation frameworks for simulation kernels [Gui et al. \[2020\]](#), [Li et al. \[2022\]](#), [Jin et al. \[2024\]](#), [Cowtan et al. \[2020b\]](#) reorder or regroup Pauli terms already determined by a Trotter formula, as do term-ordering arguments on the Trotter side [Hastings et al. \[2015\]](#), [Campbell \[2019\]](#); and synthesis under connectivity constraints [Nash et al. \[2020\]](#), [Kissinger and Meijer-van de Griend \[2020\]](#), [Gheorghiu et al. \[2023\]](#) realises a given linear-reversible or phase-polynomial block on the graph — a line of work with a ZX-calculus reading [van de Wetering \[2020\]](#) and a counterpart in swap-network constructions that engineer connectivity into a *fixed* template [Parella-Dilmé et al. \[2026\]](#), [Cheng et al. \[2022\]](#), [Kivlichan et al. \[2018\]](#). Bounds on routing as token swapping on graphs [Childs et al. \[2019\]](#), ion-trap-specific compilation [Maslov \[2017\]](#), and the production transpiler stacks [Javadi-Abhari et al. \[2024\]](#) all share the same premise. In all of these the set of two-qubit interactions to be realised is an input. Ours is an output.

On the algorithm side, the ADAPT-VQE family [Grimsley et al. \[2019\]](#), [Tang et al. \[2021\]](#), [Ryabinkin et al. \[2018, 2020\]](#) scores candidate generators by an energy gradient alone; “hardware-efficient” in qubit-ADAPT-VQE [Tang et al. \[2021\]](#) means cheap *generators*, not device topology, and later refinements shape circuit *form* — TETRIS-ADAPT-VQE packs disjoint operators into layers [Anastasiou et al. \[2024\]](#), and its dynamics counterpart AVQDS(T) [Zhang et al. \[2025\]](#) does the same for AVQDS and additionally prefers operators on idle qubits — without any coupling graph entering the criterion. The same is true of refinements that change the growth *target* or the pool’s construction rather than its cost — overlap-guided selection [Feniou et al. \[2023\]](#), symmetry-aware and measurement-frugal pools [Shkolnikov et al. \[2023\]](#), and operator-pool tiling for scale [Van Dyke et al. \[2024\]](#). All are orthogonal to, and composable with, what we do.

The closest prior work is contemporaneous and on the ground-state side: Co-ADAPT-VQE [Ramôa et al. \[2026\]](#) divides the ADAPT energy gradient by a hardware penalty (transpiled CNOT count, gate

error, or coherence time) and tracks a qubit assignment that SWAP or FSWAP networks update during growth; and Ref. [Umer and Angelakis \[2026\]](#) restricts an adaptive pool to physically connected pairs. Our contribution is distinct on three axes. (i) *Setting and functional*. Theirs optimise a static energy, where the gradient is a heuristic progress proxy carrying no accuracy guarantee; ours is real-time dynamics, where L^2 is a certified tracking-error functional and the algorithm’s correctness rests on the invariant $L^2 < L_{\text{cut}}^2$ holding at every step. (ii) *Form of the rule*. A penalised scalarisation g_ν/p_ν lets hardware cost outvote the physics criterion, and can select a generator that fails it. We instead form an *acceptability set* from McLachlan’s own criterion and take a pure argmin of routing cost *within* it, so hardware cost is a strict secondary objective and the accuracy guarantee survives by construction — constrained lexicographic selection rather than penalised scalarisation. Restricting the pool outright [Umer and Angelakis \[2026\]](#) is the opposite extreme: on our Topology B it would delete three of the eight Ising bonds under every placement, leaving growth events with no operator able to reach L_{cut}^2 . (iii) *Baseline*. Ref. [Ramôa et al. \[2026\]](#) compares against vanilla ADAPT-VQE; we compare against unconstrained AVQDS *plus* an exhaustively optimised placement *plus* post-hoc shortest-path SWAP routing, using the identical router on both sides, which isolates routing-aware *growth* from routing-aware *compilation*. Section 5.1 shows the margin comes from the operator sequence, not from routing cleverness.

The paper is organised as follows. Section 2 states the method in full, including the cost model, the acceptance rule, the online router, and the placement rule, all before any results. Section 3 fixes the benchmarks, topologies and verification protocol. Section 4 reports the measurements. Section 5 analyses the mechanism, the failure modes — the method’s depth advantage is much smaller than its CNOT advantage, and on one benchmark it is negative — and the computational cost. Section 6 concludes.

2 Cost-regularised ansatz growth

This section specifies the method completely. Section 2.1 fixes notation and recalls the growth step we modify; 2.2 identifies the freedom we exploit; 2.3 defines the hardware cost oracle; 2.4 states the acceptance rule; 2.5 the online router; 2.6 the placement rule; 2.7 the software separation and the hyperparameters. No empirical result is used anywhere in this section.

2.1 The AVQDS growth step

Let $|\Psi[\theta]\rangle$ be the pseudo-Trotter state of Eq. (1), with $\hat{A}_\mu$ Pauli strings from a pool $\mathcal{P}$. McLachlan’s variational principle [McLachlan \[1964\]](#), [Yuan et al. \[2019\]](#) minimises, over θ , the distance between the variational velocity and the exact one,

$$L^2(\dot{\theta}) = \left\| \sum_\mu \frac{\partial \rho}{\partial \theta_\mu} \dot{\theta}_\mu + i[\hat{\mathcal{H}}, \rho] \right\|_F^2 = \sum_{\mu\nu} M_{\mu\nu} \dot{\theta}_\mu \dot{\theta}_\nu - 2 \sum_\mu V_\mu \dot{\theta}_\mu + 2(\langle \hat{\mathcal{H}}^2 \rangle - \langle \hat{\mathcal{H}} \rangle^2), \quad (2)$$

with $\rho = |\Psi\rangle\langle\Psi|$, M the (real, symmetric) quantum Fisher information matrix and V the force vector, both defined in Ref. [Yao et al. \[2021\]](#). The minimiser is $\dot{\theta} = (M + \xi\mathbb{K})^{-1}V$ with Tikhonov regularisation ξ , and the attained residual is, up to an overall factor of two that we drop with Ref. [Yao et al. \[2021\]](#),

$$L^2 = \underbrace{\langle \hat{\mathcal{H}}^2 \rangle - \langle \hat{\mathcal{H}} \rangle^2}_{\equiv \sigma_{\mathcal{H}}^2} - \underbrace{V^\top (M + \xi\mathbb{K})^{-1} V}_{\equiv \Pi}. \quad (3)$$

Only Π depends on the ansatz; $\sigma_{\mathcal{H}}^2$ is a property of the current state. Appending a trial operator $\hat{A}_\nu$ with $\theta_\nu = 0$ leaves $|\Psi\rangle$ (hence $\sigma_{\mathcal{H}}^2$) unchanged and borders M and V by one row and column, giving $\Pi_\nu \geq \Pi$ and $L_\nu^2 = \sigma_{\mathcal{H}}^2 - \Pi_\nu$. AVQDS evaluates $(\Pi_\nu, \|\dot{\theta}_\nu\|_\infty)$ for every $\nu \in \mathcal{P}$, commits one operator, and repeats until $L^2 < L_{\text{cut}}^2$. The time step is then $\delta t = \min(\delta t_{\text{max}}, \delta \theta_{\text{max}} / \|\dot{\theta}\|_\infty)$ and $\theta \leftarrow \theta + \delta t \dot{\theta}$ (forward Euler).

We keep every element of this description. Our modification concerns only the words “commits one operator”.

2.2 The freedom in the growth step

Write $\Delta_\nu = \Pi_\nu - \Pi \geq 0$ for the improvement a candidate brings. Two structural facts make the argmax over Δ_ν an over-determination.

(i) *The stopping criterion is an inequality.* Whenever some candidate satisfies $L_\nu^2 < L_{\text{cut}}^2$, the growth loop terminates after committing it. Every candidate in the *sufficient set* $\mathcal{S} = \{\nu : L_\nu^2 < L_{\text{cut}}^2 \text{ and } \|\hat{\theta}_\nu\|_\infty < \vartheta_{\text{max}}\}$ therefore discharges the algorithm’s accuracy obligation identically, and the algorithm’s preference among them is a free choice. The reference implementation already exercises that freedom, breaking the tie by conditioning (it prefers the candidate with the smallest $\|\hat{\theta}_\nu\|_\infty$). Our rule occupies exactly this slot.

(ii) *Near-degeneracy of $\{L_\nu^2\}$.* When no single operator suffices, the growth loop must add several, and the Δ_ν are frequently almost degenerate across the pool — an observation reported in Ref. Yao et al. [2021] and used there to explain why an enlarged pool produces *longer* circuits without an accuracy gain. Committing the ε -suboptimal operator instead of the optimal one moves the ansatz along a different path through operator space. Whether that path needs more insertions is an empirical question, not something the near-degeneracy argument settles, and the answer turns out to be yes: our ansätze carry $1.42\times$ the parameters of the unconstrained ones on Topology B. That is the price of the freedom, and Sec. 5.2 treats it as a first-class cost rather than a rounding error, because the AVQDS measurement budget grows as N_θ^2 .

The design question is therefore not *whether* there is slack, but what to spend it on. We spend it on the device graph.

2.3 Hardware cost oracle

Let $G = (Q, E)$ be the connectivity graph, d_G its shortest-path metric, and let σ be the current *placement*, a bijection from logical spin sites to physical qubits ($\sigma(s)$ is the qubit currently holding site s). Placements change as SWAPs are inserted; σ at a given growth event is whatever the online router (Sec. 2.5) has reached.

We use the same gate-level conventions for every method compared in this paper. A weight-1 Pauli rotation compiles to single-qubit gates only. A weight-2 Pauli rotation on an *adjacent* physical pair compiles to {basis change, CNOT, R_z , CNOT, basis change}, i.e. exactly 2 CNOTs — the all-to-all convention of Ref. Yao et al. [2021]. A SWAP costs 3 CNOTs — the optimal count for a generic two-qubit unitary Vidal and Dawson [2004], Vatan and Williams [2004], Shende et al. [2006], and the standard decomposition into elementary gates Barenco et al. [1995]. A rotation whose endpoints are at distance $d > 1$ is realised by first moving them together with $d - 1$ SWAPs along a shortest path and then applying the 2-CNOT rotation. Hence the cost oracle

$$c_\sigma(\hat{A}_\nu) = \begin{cases} 0, & \hat{A}_\nu \text{ of weight 1,} \\ 3 d_G(\sigma(i_\nu), \sigma(j_\nu)) - 1, & \hat{A}_\nu \text{ of weight 2 on sites } (i_\nu, j_\nu), \end{cases} \quad (4)$$

which returns 2 for an adjacent pair, so the oracle is continuous with the unconstrained convention and no term is hidden. Evaluating Eq. (4) is one lookup in a precomputed $|Q| \times |Q|$ distance matrix.

Two remarks. First, c_σ is a *myopic* cost: it prices the operator as if it were routed now and the placement never used again. It ignores the fact that the SWAPs it pays for also improve (or worsen) the cost of later gates. We deliberately keep it myopic — the amortisation is handled by the router’s lookahead, not by the selection rule — because a myopic oracle makes the acceptance rule a pure function of the current state and keeps the contribution auditable. Second, c_σ assigns cost 0 to single-qubit generators. This is physically correct under the CNOT metric, and it means our rule prefers a single-qubit rotation whenever one is acceptable. That preference is *topology-independent*, and we isolate it by ablation in Sec. 5.1 rather than folding it into the topology claim.

2.4 The acceptance rule

Let $\mathcal{F} = \{\nu : \Delta_\nu > \epsilon_0\}$ be the feasible set of candidates that make progress ($\epsilon_0 = 10^{-8}$, the reference implementation’s own threshold), $\Delta_{\text{max}} = \max_{\nu \in \mathcal{F}} \Delta_\nu$, and $\mathcal{S} \subseteq \mathcal{F}$ the sufficient set of

Sec. 2.2. Given a tolerance band $\eta \in [0, 1)$, define the candidate set

$$\mathcal{C}_\eta = \begin{cases} \mathcal{S}, & \mathcal{S} \neq \emptyset \text{ (cheapest sufficient operator),} \\ \{\nu \in \mathcal{F} : \Delta_\nu \geq (1 - \eta) \Delta_{\max}\}, & \mathcal{S} = \emptyset \text{ (tolerance band),} \end{cases} \quad (5)$$

and commit

$$\boxed{\nu^* = \arg \min_{\nu \in \mathcal{C}_\eta} \left(c_\sigma(\hat{A}_\nu), \|\dot{\theta}_\nu\|_\infty, -\Delta_\nu, \nu \right)} \quad (6)$$

lexicographically. The first key is the hardware cost; the second is the reference implementation’s own conditioning heuristic, retained as a tie-breaker so that on a topology where all pool operators cost the same we degrade gracefully to the original secondary rule; the third prefers the larger EOM improvement; the fourth is the pool index, making the rule deterministic.

The band is scale-free: $\Delta_\nu \geq (1 - \eta) \Delta_{\max}$ compares improvements to the best available improvement, not to an absolute scale, so a single η is meaningful across benchmarks and across time. Setting $\eta = 0$ leaves only the first branch of Eq. (5) active, i.e. “cheapest sufficient operator, otherwise plain AVQDS”.

Guarantees. (a) *Accuracy is not traded away.* Every $\nu \in \mathcal{C}_\eta$ lies in $\mathcal{F}$, so Π strictly increases at every commit and the growth loop still exits only when $L^2 < L_{\text{cut}}^2$; the threshold that controls AVQDS’s accuracy is enforced verbatim. What changes is which operators are used to satisfy it, and possibly how many. (b) *Termination.* If $\mathcal{F} = \emptyset$ we fall back to the reference rule, which raises the same exception AVQDS would; likewise if the selected candidate would improve neither Π (by more than 10^{-8}) nor the conditioning, we defer to the reference choice — this is the reference implementation’s own stall guard, kept verbatim. To bound the pathological case in which repeatedly choosing a cheap operator stalls the loop, we also revert to the reference choice once a single growth event has run $\kappa/2$ inner cycles, and abort beyond κ ; with $\kappa = 60$ neither ever fired in the runs reported here (Sec. 5.1). (c) *Exact reduction.* With $\eta = 0$ and $c_\sigma \equiv \text{const}$, Eq. (6) reduces to the reference implementation’s streaming rule up to tie-breaking, and our implementation additionally routes the topology $= \emptyset$ case through the reference rule itself, so the unconstrained baseline is the original algorithm exactly (Sec. 2.7).

2.5 Online router

The router maintains σ and emits gates causally as operators are committed. On committing a weight-2 operator on sites (i, j) with $u = \sigma(i)$, $v = \sigma(j)$ and $d = d_G(u, v)$: if $d = 1$ it emits the 2-CNOT rotation. Otherwise it takes a shortest path $u = p_0, \dots, p_d = v$ and must place $d - 1$ SWAPs on it; the only freedom is how many advance u and how many retract v , giving d splittings. It scores each by the resulting placement’s potential

$$\Phi(\sigma') = \sum_{(i,j)} w_{ij} d_G(\sigma'(i), \sigma'(j)), \quad (7)$$

picks the minimiser, emits the SWAPs, updates σ , and then emits the rotation. SWAPs are never undone: the placement is left where the routing put it, so a SWAP paid for once benefits all later gates.

The weights w_{ij} define the lookahead and are the one place where the online and the post-hoc settings differ, necessarily so:

- *Online* (inside CRAG-AVQDS, where the future is not yet written): w_{ij} is an exponentially weighted moving average of how often the pool operator on sites (i, j) has recently been committed, updated by $w \leftarrow 0.98 w$ each two-qubit commit with a unit increment on the committed pair. This is a *prediction* of future demand from realised demand, available because the pool is small and its usage is locally stationary.
- *Post-hoc* (applied to a completed operator sequence): w_{ij} is a geometrically decayed count over the next W operators of the known remaining sequence, $w \propto \lambda^k$ for the k -th upcoming operator — the standard SABRE-style lookahead Li et al. [2019].

Algorithm 1 CRAG-AVQDS growth event (replaces the commit rule; everything else is the AVQDS core of Ref. Yao et al. [2021]).

Require: pool $\mathcal{P}$, threshold L_{cut}^2 , conditioning cap ϑ_{max} , band η , graph G , router state σ

- 1: **repeat**
- 2: **for** $\nu \in \mathcal{P}$ with $\hat{A}_\nu \neq$ last committed operator **do**
- 3: border M, V by $\hat{A}_\nu$; $\Pi_\nu \leftarrow V^\top (M + \xi \mathbb{K})^{-1} V$; $\|\dot{\theta}_\nu\|_\infty \leftarrow \|(M + \xi \mathbb{K})^{-1} V\|_\infty$ $\triangleright$
 authors' code, unchanged
- 4: **end for**
- 5: $\mathcal{F} \leftarrow \{\nu : \Pi_\nu - \Pi > 10^{-8}\}$; $\mathcal{S} \leftarrow \{\nu \in \mathcal{F} : \sigma_{\mathcal{H}}^2 - \Pi_\nu < L_{\text{cut}}^2 \wedge \|\dot{\theta}_\nu\|_\infty < \vartheta_{\text{max}}\}$
- 6: $\mathcal{C} \leftarrow \mathcal{S}$ **if** $\mathcal{S} \neq \emptyset$ **else** $\{\nu \in \mathcal{F} : \Pi_\nu - \Pi \geq (1 - \eta) \max_{\mathcal{F}} (\Pi - \Pi)\}$
- 7: $\nu^* \leftarrow \arg \min_{\nu \in \mathcal{C}} (c_\sigma(\hat{A}_\nu), \|\dot{\theta}_\nu\|_\infty, -\Delta_\nu, \nu)$ $\triangleright$ the contribution
- 8: commit $\hat{A}_{\nu^*}$ with $\theta = 0$; route it (Sec. 2.5), updating σ ; $\Pi \leftarrow \Pi_{\nu^*}$
- 9: **until** $\sigma_{\mathcal{H}}^2 - \Pi < L_{\text{cut}}^2$ **and** $\|\dot{\theta}\|_\infty < \vartheta_{\text{max}}$

The post-hoc router with known future is strictly better informed. We therefore apply it, with an identical configuration search, to *both* the unconstrained sequence (producing the naive baseline) and our own sequence, and report the best of $\{\text{no lookahead, usage lookahead, } (W, \lambda) \in \{(5, 0.8), (20, 0.5), (20, 0.9), (200, 0.98)\}\}$ in each case. Any margin between the two therefore comes from the operator sequence, not from the router.

2.6 Placement rule

The initial placement is fixed *a priori* from the Hamiltonian's interaction graph alone — no dynamics is run — by exhaustive enumeration of all $8! = 40320$ bijections, minimising lexicographically

$$\left(\sum_{b \in \mathcal{B}} d_G(\sigma(i_b), \sigma(j_b)), \quad |\{b \in \mathcal{B} : d_G(\sigma(i_b), \sigma(j_b)) > 1\}|, \quad \sigma \text{ lexicographically} \right), \quad (8)$$

where $\mathcal{B}$ is the set of two-site Hamiltonian bonds. The first key is total routing pressure, the second concentrates the residual pressure on as few bonds as possible, the third makes the choice deterministic. *The same placement is given to the naive baseline and to CRAG-AVQDS.*

Because a favourable placement could otherwise be mistaken for a favourable algorithm, we additionally grant the naive baseline an *oracle* placement: the bijection, among all $8!$, that minimises the baseline's own realised routed CNOT count. It is found by scoring all 40320 placements with the cheapest router and re-routing the best 128 — together with the a-priori placement itself — under every router configuration; including the a-priori placement in that shortlist guarantees the oracle result is never worse than the a-priori one, so this search can only help the baseline. (We report whichever of the two is better, so the “naive*” column is by construction a lower bound on what placement search alone can achieve within our router.) CRAG-AVQDS receives no such advantage; instead we report its spread over an unbiased sample of the placements tied at the optimum of Eq. (8) (Sec. 4), so that the sensitivity of our number to the placement choice is visible.

2.7 Implementation, hyperparameters and separation of concerns

Algorithm 1 shows the growth event. The implementation is a subclass of the reference ansatz class overriding one method, `add_ops_dyn`. The per-candidate linear algebra inside the override is a line-for-line transcription of the reference code (the original fuses its accept test into the candidate loop as a streaming update, leaving no hook to intercept); the override *also* evaluates the original streaming rule on the same candidate table, and when the topology layer is disabled it returns that choice. This gives the single switch the design calls for: with `topology =  $\emptyset$` and $\eta = 0$ the subclass reproduces a full reference trajectory bit-for-bit — identical step count, identical operator sequence, $\max |\Delta \theta| = 0$, $\max |\Delta \Psi| = 0$ (test code/tests/test_memoisation.py). The cost oracle, the router, the placement search and the compiler live in separate modules (`topology.py`, `compiler.py`, `embeddings.py`) that the AVQDS core never calls.

Hyperparameters. AVQDS parameters are taken from Ref. Yao et al. [2021] and its reference scripts and are identical for every method: $L_{\text{cut}}^2 = 10^{-3}$, $\xi = 10^{-6}$, $\delta \theta_{\text{max}} = 5 \times 10^{-3}$, $\delta t_{\text{max}} = 0.05$,

$\vartheta_{\max} = 7000$. The topology layer adds one hyperparameter, η , selected by a rule fixed in advance: sweep $\eta \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5\}$ on a single designated tuning configuration — MFIM $h_z = 0$ on Topology B — keep the values whose final-time fidelity is within the 1% tolerance of the unconstrained baseline, and among those take the smallest CNOT multiplier, breaking ties towards smaller η . The selected value is then held fixed for every other benchmark and every other topology, so eleven of the twelve reported (benchmark, topology) cells are out-of-sample with respect to the hyperparameter. We nevertheless run the complete grid on the other benchmarks as well and report it (Appendix Table 4), so the reader can see what a per-benchmark choice would have given. The inner-cycle cap is $\kappa = 60$. The router’s EWMA decay 0.98 and the post-hoc lookahead grid are shared by all methods.

What we do not claim. The compilation primitives are standard: SWAP networks, shortest-path routing and SABRE-style lookahead are prior art [Li et al. \[2019\]](#), [Cowtan et al. \[2019\]](#), [Zulehner et al. \[2019\]](#). We do not propose a new router, and we deliberately give the baseline the same one. Neither do we propose a new variational principle: Eq. (3) and its integration are Ref. [Yao et al. \[2021\]](#) unchanged, and the underlying product-formula machinery against which such methods are benchmarked [Trotter \[1959\]](#), [Suzuki \[1990, 1991\]](#) is untouched. The claim is confined to Eq. (6) — that the acceptance slack inside an adaptive growth step is a usable resource for hardware compliance, and that using it is worth more than routing the unconstrained circuit optimally.

3 Benchmarks, topologies and verification protocol

3.1 Models

Both benchmarks are eight-qubit spin chains, simulated by noiseless dense statevector propagation.

Mixed-field Ising model (MFIM), sudden quench.

$$\hat{\mathcal{H}}_{\text{MFIM}} = -J \sum_{i=0}^{N-1} \hat{Z}_i \hat{Z}_{i+1} + \sum_{i=0}^{N-1} (h_x \hat{X}_i + h_z \hat{Z}_i), \quad \hat{Z}_N \equiv \hat{Z}_0, \quad (9)$$

with $N = 8$, periodic boundaries, $J = 1$, $h_x = -2$ and $h_z \in \{0, 0.5\}$ ($h_z = 0$ is the integrable transverse-field limit, $h_z = 0.5$ is nonintegrable). The system starts in $|\Psi_0\rangle = |0\rangle^{\otimes 8}$, the ground state of the field-free Hamiltonian, and the full Hamiltonian is switched on at $t = 0$. Final time $T = 3$ in units of $1/J$. The Hamiltonian pool is $\{\hat{X}_i\}_{i=0}^7 \cup \{\hat{Z}_i \hat{Z}_{i+1}\}_{i=0}^7$ (16 operators) at $h_z = 0$, with $\{\hat{Z}_i\}_{i=0}^7$ added (24 operators) at $h_z = 0.5$. The eight ZZ bonds, including the periodic bond $\hat{Z}_7 \hat{Z}_0$, form a ring.

Lieb–Schultz–Mattis (LSM) chain, finite-rate ramp.

$$\hat{\mathcal{H}}_{\text{LSM}}(t) = -J \sum_{i=0}^{N-2} \left[(1 + \gamma(t)) \hat{X}_i \hat{X}_{i+1} + (1 - \gamma(t)) \hat{Y}_i \hat{Y}_{i+1} \right] + h_z \sum_{i=0}^{N-1} \hat{Z}_i, \quad (10)$$

with $N = 8$, open boundaries, $J = 1$, and the linear ramp $\gamma(t) = 1 - 2t/T$ for $0 \leq t \leq T$, $T = 3$, driving the system from the FM_x phase across the transition into FM_y [Lieb et al. \[1961\]](#); a finite-rate crossing of a critical point in the Kibble–Zurek sense [Kibble \[1976\]](#), [Zurek \[1985\]](#), for $t > T$ the Hamiltonian is held at $\hat{\mathcal{H}}_{\text{LSM}}(T)$ and the post-ramp dynamics is followed for a further period T , so the total simulated time is $2T = 6$. The two quench paths are $h_z = -0.7$ and $h_z = 1.6$, chosen (as in Ref. [Yao et al. \[2021\]](#)) to avoid ground-state degeneracy. The initial state is the exact ground state of $\hat{\mathcal{H}}_{\text{LSM}}(0)$ from classical diagonalisation, taken as the reference state $|\Psi_0\rangle$ of the ansatz. The Hamiltonian pool is $\{\hat{Z}_i\}_{i=0}^7 \cup \{\hat{X}_i \hat{X}_{i+1}\}_{i=0}^6 \cup \{\hat{Y}_i \hat{Y}_{i+1}\}_{i=0}^6$ (22 operators); the seven bonds form an open path, each carrying two pool operators.

Both pools are the canonical Hamiltonian pools generated by the reference implementation; no expanded pool is used anywhere in this paper. Operators and states are represented with QuTiP [Johansson et al. \[2013\]](#), as in the reference implementation.

MFIM $h_z = 0$: spin site (inside node) $\rightarrow$ physical qubit (grey label), and realised two-qubit gate traffic

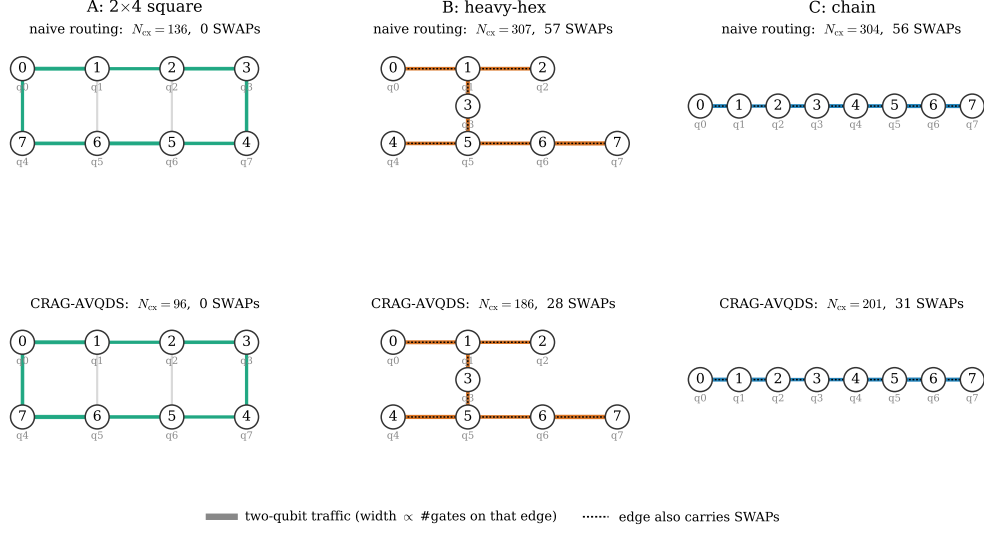


Figure 1: The three connectivity graphs, the placement chosen by Eq. (8), and the two-qubit traffic each method actually generates on MFIM at $h_z = 0$. Node interiors give the logical spin site, the small label below gives the physical qubit. Edge thickness is proportional to the number of two-qubit operations executed on that edge; a dotted overlay marks edges that also carry SWAPs. Top row: smart-embedding naive routing of the unconstrained circuit. Bottom row: CRAG-AVQDS.

3.2 Topologies

Three graphs on eight qubits are used (Fig. 1). The first two are the pair we study throughout; the third tests whether the method transfers without retuning.

A — 2×4 square lattice (10 edges). Rows $0 - 1 - 2 - 3$ and $4 - 5 - 6 - 7$ with rungs $(0, 4), (1, 5), (2, 6), (3, 7)$. Degree sequence $(3, 3, 3, 3, 2, 2, 2, 2)$. It admits the Hamiltonian cycle $0 - 1 - 2 - 3 - 7 - 6 - 5 - 4 - 0$, so the MFIM ring embeds with all eight bonds at distance one, and *a fortiori* so does the LSM path.

B — eight-qubit heavy-hex subgraph (7 edges). A representative patch of IBM’s heavy-hex architecture Chamberland et al. [2020], Kim et al. [2023], with edge set $\{(0, 1), (1, 2), (1, 3), (3, 5), (4, 5), (5, 6), (6, 7)\}$. With 8 vertices and 7 edges it is a *tree*, with four leaves $\{0, 2, 4, 7\}$. A Hamiltonian path has at most two degree-one endpoints, so none exists; a tree admits a Hamiltonian path only if it is a path. Consequently neither the MFIM ring nor the LSM chain can be embedded locally under any placement. For the ring the bound is sharper: the eight bonds trace a closed walk that must cross every tree edge an even number of times, hence $\sum_b d_G \geq 2 \cdot 7 = 14$, and exhaustive enumeration confirms 14 is attained (with at least three bonds non-local, distances $\{1, 1, 1, 1, 2, 2, 5\}$). For the LSM path exhaustive enumeration gives $\min \sum_b d_G = 9$ with two bonds at distance 2.

C — linear chain (7 edges). $0 - 1 - \dots - 7$. The LSM path embeds perfectly; the MFIM ring cannot, and its optimum is one single bond at distance 7 with the other seven local ($\sum_b d_G = 14$, the same total as B but concentrated on one bond instead of three). C is thus a controlled contrast to B at equal routing pressure.

Placements follow Eq. (8); the resulting distance profiles are listed above and logged in `logs/runs/*.json` under `embedding_info`.

3.3 Comparison methods

Unconstrained AVQDS. The reference implementation of Ref. Yao et al. [2021], run through our integration loop, compiled under full connectivity: 2 CNOTs per two-qubit Pauli rotation. This is the denominator of every overhead multiplier.

Smart-embedding naive routing. The *same* operator sequence, placed by Eq. (8) and routed post hoc by the shared router of Sec. 2.5 with the best of its six configurations. This is the reference for “good routing without ansatz adaptation”.

Naive* (oracle placement). Identical, but with the placement that minimises its own realised CNOT count over all $8!$ bijections, taken as the better of that search and the a-priori placement (Sec. 2.6). This is a strictly stronger baseline than the protocol requires, and it is the one we compare against in the abstract and the conclusion.

CRAG-AVQDS. Our method, placed by Eq. (8) and compiled with the same shared post-hoc router and the same configuration search.

Exact reference. For MFIM, $e^{-i\hat{\mathcal{H}}t} |\Psi_0\rangle$ from the eigendecomposition of $\hat{\mathcal{H}}$ (numerically exact at every time). For LSM, integration of the ramp on a uniform mesh $\delta t_{\text{exact}} = 5 \times 10^{-4}$ with the mid-point Hamiltonian and a Taylor propagator converged to machine precision, matching the reference protocol of Ref. Yao et al. [2021]; a query at an arbitrary time is covered by one further sub-mesh Taylor step.

Within a benchmark, all methods share the initial state and, where they share an operator sequence (unconstrained and both naive variants), the identical time-step schedule. CRAG-AVQDS runs its own trajectory — it must, since its ansatz differs — but with identical $\delta\theta_{\text{max}}$, δt_{max} , L_{cut}^2 , ξ and ϑ_{max} , so the step schedule is generated by the same rule.

3.4 Metrics

Fidelity $f(t) = |\langle \Psi_{\text{exact}}(t) | \Psi[\theta(t)] \rangle|^2$; CNOT count N_{cx} counting *every* CNOT, including the three inside each SWAP; circuit depth, defined as the ASAP-scheduled depth of the compiled primitive gate list over $\{1\text{-qubit gates, CNOT}\}$, with SWAPs expanded into their three CNOTs (we also log CNOT-only depth); number of variational parameters N_{θ} ; McLachlan distance L^2 ; Loschmidt echo $|\langle \Psi_0 | \Psi(t) \rangle|^2$ (MFIM); energy $\langle \hat{\mathcal{H}}(t) \rangle$ and the correlators $\langle \sigma_0^x \sigma_1^x \rangle$, $\langle \sigma_0^x \sigma_7^x \rangle$, $\langle \sigma_0^y \sigma_1^y \rangle$, $\langle \sigma_0^y \sigma_7^y \rangle$ (LSM). All are recorded at every time step. The depth definition is identical for all methods, so multipliers are meaningful even though the absolute depth depends on the convention.

3.5 Verification

Four checks run automatically over every run and every circuit reported.

(1) *Unitary equivalence.* An independent statevector simulator, written against the primitive gate list rather than the Pauli-exponential representation, executes the compiled circuit on $|\Psi_0\rangle$ placed by σ_{initial} , undoes the final tracked placement, and compares against the state the AVQDS core produced. Across all 40 compiled circuits reported here, the largest amplitude deviation is 6.8×10^{-15} . This certifies the compilation, the SWAP bookkeeping and the placement tracking simultaneously; in particular a routing bug cannot silently inflate or deflate a gate count without breaking this check. The residual placement at the end of a circuit is a classical relabelling of measurement outcomes and costs no gates; it is applied identically to every method.

(2) *Topology compliance.* Every two-qubit primitive in every circuit is checked against the relevant edge set, and the per-pair histogram is written to the run record (`circuits.*.audit`), with per-topology gate-level listings in `logs/compliance_audit_{A,B,C}.json`. Across all reported topology-aware and naive circuits the number of violations is 0. For the naive baseline specifically, `logs/naive_routing_side_by_side.txt` places the unconstrained source circuit and the SWAP-inserted circuit next to each other — the source rotations with their graph distances under the placement, then the routed two-qubit stream in circuit order with an edge-membership flag on each entry — so the post-hoc transformation can be audited gate by gate. Because routing decisions depend only on the operator sequence, the placement and the router configuration, that listing is regenerated from the stored run record and cross-checked against the CNOT count logged at run time.

(3) *Ansatz-growth integrity.* The growth decision uses only M , V , $\sigma_{\mathcal{H}}^2$ and the graph distance — all classically measurable without reference to $|\Psi_{\text{exact}}\rangle$. The exact state is computed only to *report* fidelity and never enters Eqs. (5)–(6); the code paths are separate modules.

(4) *Initial-state and schedule pairing.* Every run of a benchmark is checked against an initial state recomputed independently from the benchmark definition ($|0\rangle^{\otimes 8}$ for MFIM, the diagonalised ground state of $\hat{\mathcal{H}}(0)$ for LSM); the largest deviation over all runs is 8×10^{-17} , and every run of a benchmark uses one identical hyperparameter set, verified field by field (`logs/pairing_audit.json`). The naive baselines share the unconstrained run’s trajectory exactly, being post-hoc recompilations of its own operator sequence and parameters stored in the same record; CRAG-AVQDS necessarily has its own trajectory, generated by the identical step-size rule.

Separately from the production runs, a randomised stress test (`code/tests/test_router.py`) subjects the compiler and router to 684 checks over random Pauli-rotation sequences, random placements, all three topologies and every router configuration, asserting unitary equivalence to the uncompiled Pauli-exponential product, edge-set compliance, the exact CNOT accounting $N_{\text{cx}} = 2n_{\text{rot}2} + 3n_{\text{swap}}$, and — for the causal router configurations — that routing a prefix of a sequence equals the corresponding prefix of routing the whole sequence, which is what makes the $N_{\text{cx}}(t)$ curves of Fig. 4 well defined. All 684 pass.

4 Results

All numbers below come from the 52 logged runs in `proposal/logs/runs/`; the tables and every number quoted in the text are generated mechanically from those records (`proposal/code/aggregate.py`, `macros.py`).

4.1 Unconstrained AVQDS reproduces the reference behaviour

Table 1 (first row) gives the unconstrained baseline. On MFIM the final-time fidelity is 99.96% at $h_z = 0$ and 99.92% at $h_z = 0.5$, and the minimum over the whole trajectory is 99.92% — comfortably above the 99.5% quoted in Ref. Yao et al. [2021]. The circuits reach $N_{\text{cx}} = 136$ ($h_z = 0$) and 230 ($h_z = 0.5$) logical CNOTs, against 134 and 210 reported there — 1% above and 10% above the published values respectively. Exact agreement is not to be expected: the pool scan has near-degenerate minima whose ordering depends on floating-point details and on $\delta\theta_{\text{max}}$, which we set to 5×10^{-3} following the paper’s text while the released run scripts use 10^{-2} . The qualitative feature that matters does reproduce — the integrable case is markedly cheaper than the nonintegrable one — and the absolute counts are close enough that the multipliers we report are meaningful relative to the published reference point. On LSM the ansatz reaches 58 and 61 two-qubit rotations at $2T = 6$ for $h_z = -0.7$ and $h_z = 1.6$ respectively, i.e. 116 and 122 CNOTs, against the ~ 100 CNOTs (50 two-qubit rotations at $t = T$, 53 at $t = 2T$) of Ref. Yao et al. [2021]; the energy and correlation trajectories (Fig. 6) track the exact ones over the full ramp and post-ramp window, including the maximum of the energy near $\gamma = 0$ and the slow growth of the long-range $\langle \sigma_0^y \sigma_7^y \rangle$ correlation, exactly as in the original Fig. 4. We conclude the baseline is configured correctly, and use it as the denominator throughout.

4.2 Selecting the tolerance band

Figure 2 (left, middle) shows the sweep over $\eta \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5\}$ on Topology B, for the designated tuning configuration and for the other benchmarks on which we ran the grid as a sensitivity check. Two things are visible. First, the accuracy cost of the band is negligible over the whole range: the largest fidelity difference from the unconstrained baseline anywhere in the sweep stays orders of magnitude below the 1% floor, because the growth loop enforces $L^2 < L_{\text{cut}}^2$ regardless of which acceptable operator is committed. The band is therefore not an accuracy–cost trade-off in the usual sense; it is a search-freedom knob. Second, the CNOT multiplier falls from $\eta = 0$ and then flattens, which is what one expects if the useful slack is concentrated in near-degenerate candidates. Applying the pre-registered selection rule of Sec. 2.7 to the tuning configuration gives $\eta = 0.20$, and this single value is used for every reported CRAG-AVQDS run, on every benchmark and every topology.

Note that $\eta = 0$ is *not* the unconstrained algorithm: it still applies the “cheapest sufficient operator” branch of Eq. (5). That branch alone already brings the Topology-B multiplier to $1.19\times$ (averaged

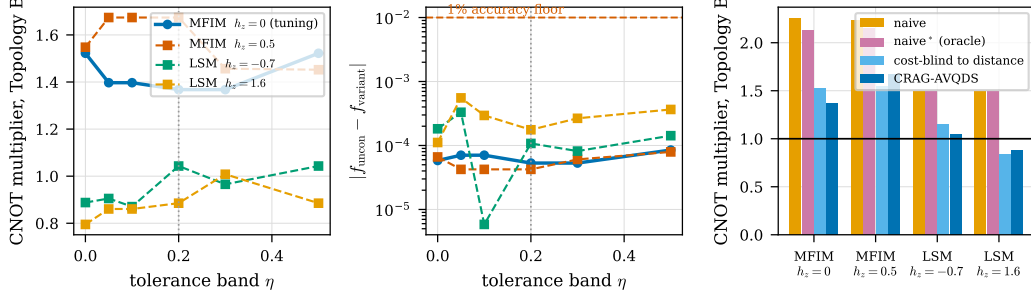


Figure 2: **Left, middle:** the complete η sweep on Topology B, for the 4 benchmarks on which the whole grid was run (MFIM $h_z = 0$, MFIM $h_z = 0.5$, LSM $h_z = -0.7$, LSM $h_z = 1.6$); all six values, nothing withheld. The tuning configuration is drawn with a solid line, the sensitivity check dashed. Left, CNOT multiplier against unconstrained AVQDS; middle, the absolute fidelity difference from the unconstrained baseline against the 1% floor. The dotted vertical line marks the selected $\eta = 0.20$; note the response is not monotone and the selected value is not optimal for the dashed benchmark. **Right:** component ablation on Topology B. “cost-blind to distance” keeps the acceptance rule but flattens the cost oracle to 2 for every two-qubit rotation, isolating how much of the gain is graph-distance awareness rather than a preference for weight-one generators; on MFIM $h_z = 0.5$ it is the better of the two.

Table 1: Mixed-field Ising model at the final simulation time ($T = 3$). N_{cx} counts every CNOT, including the three CNOTs of each inserted SWAP. Depth is the ASAP depth of the compiled circuit over $\{1\text{-qubit gates, CNOT}\}$. All entries are from the logged runs listed in `proposal/logs/runs/`.

Method	Topology	MFIM $h_z = 0$			MFIM $h_z = 0.5$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
Unconstrained AVQDS	all-to-all	0.99961	136	126	0.99921	230	202
Smart-embedding naive routing	A (square)	0.99961	136	126	0.99921	230	202
Smart-embedding naive routing	B (heavy-hex)	0.99961	307	264	0.99921	515	388
Smart-embedding naive routing	C (chain)	0.99961	304	225	0.99921	485	336
naive* (oracle placement)	A (square)	0.99961	136	126	0.99921	230	202
naive* (oracle placement)	B (heavy-hex)	0.99961	289	210	0.99921	494	391
naive* (oracle placement)	C (chain)	0.99961	280	204	0.99921	455	350
CRAG-AVQDS (ours)	A (square)	0.99961	96	108	0.99922	158	172
CRAG-AVQDS (ours)	B (heavy-hex)	0.99966	186	183	0.99926	385	347
CRAG-AVQDS (ours)	C (chain)	0.99960	201	177	0.99932	343	270

over the four benchmarks), so a large part of the effect comes from exploiting the inequality in the stopping criterion rather than from tolerating suboptimal candidates.

4.3 Main results

Tables 1 and 2 give final-time fidelity, CNOT count and depth for every method on every benchmark and topology; Table 3 converts these into multipliers against unconstrained AVQDS. Figures 3 and 4 give the time resolved views.

Topology compliance. Every two-qubit primitive gate in every reported circuit lies on an edge of the relevant graph: 0 violations across all naive and topology-aware circuits on all three topologies, verified gate by gate and logged per pair (Sec. 3.5). Independently, all 40 compiled circuits reproduce the AVQDS statevector to within 6.8×10^{-15} in amplitude, so the gate counts are counts of a circuit that provably prepares the reported state.

Accuracy preservation. Across all 12 (benchmark, topology) cells the final-time fidelity of CRAG-AVQDS differs from unconstrained AVQDS by at most 0.030 % absolute (largest deviation on LSM

Table 2: Lieb–Schultz–Mattis chain at the final simulation time ($2T = 6$). N_{cx} counts every CNOT, including the three CNOTs of each inserted SWAP. Depth is the ASAP depth of the compiled circuit over $\{1\text{-qubit gates, CNOT}\}$. All entries are from the logged runs listed in `proposal/logs/runs/`.

Method	Topology	LSM $h_z = -0.7$			LSM $h_z = 1.6$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
Unconstrained AVQDS	all-to-all	0.99965	116	192	0.99957	122	189
Smart-embedding naive routing	A (square)	0.99965	116	192	0.99957	122	189
Smart-embedding naive routing	B (heavy-hex)	0.99965	173	253	0.99957	182	214
Smart-embedding naive routing	C (chain)	0.99965	116	192	0.99957	122	189
naive* (oracle placement)	A (square)	0.99965	116	192	0.99957	122	189
naive* (oracle placement)	B (heavy-hex)	0.99965	173	253	0.99957	182	214
naive* (oracle placement)	C (chain)	0.99965	116	192	0.99957	122	189
CRAG-AVQDS (ours)	A (square)	0.99959	82	105	0.99926	66	104
CRAG-AVQDS (ours)	B (heavy-hex)	0.99954	121	194	0.99939	108	170
CRAG-AVQDS (ours)	C (chain)	0.99959	82	105	0.99926	66	104

Table 3: Overhead relative to unconstrained AVQDS on the same benchmark: CNOT multiplier $N_{\text{cx}}/N_{\text{cx}}^{\text{uncon}}$ and depth multiplier. Lower is better; 1.00 means the topology constraint costs nothing. ‘naive’ is the smart-embedding post-hoc SWAP-routing baseline under the a-priori embedding rule, ‘naive*’ the same baseline given the oracle embedding that minimises its own realised CNOT count over all $8!$ placements.

Benchmark	Topology	naive		naive*		CRAG-AVQDS	
		N_{cx}	depth	N_{cx}	depth	N_{cx}	depth
MFIM $h_z = 0$	A (square)	$1.00\times$	$1.00\times$	$1.00\times$	$1.00\times$	$0.71\times$	$0.86\times$
MFIM $h_z = 0.5$	A (square)	$1.00\times$	$1.00\times$	$1.00\times$	$1.00\times$	$0.69\times$	$0.85\times$
LSM $h_z = -0.7$	A (square)	$1.00\times$	$1.00\times$	$1.00\times$	$1.00\times$	$0.71\times$	$0.55\times$
LSM $h_z = 1.6$	A (square)	$1.00\times$	$1.00\times$	$1.00\times$	$1.00\times$	$0.54\times$	$0.55\times$
<i>mean</i>	A (square)	$1.00\times$	$1.00\times$	$1.00\times$	$1.00\times$	$0.66\times$	$0.70\times$
MFIM $h_z = 0$	B (heavy-hex)	$2.26\times$	$2.10\times$	$2.12\times$	$1.67\times$	$1.37\times$	$1.45\times$
MFIM $h_z = 0.5$	B (heavy-hex)	$2.24\times$	$1.92\times$	$2.15\times$	$1.94\times$	$1.67\times$	$1.72\times$
LSM $h_z = -0.7$	B (heavy-hex)	$1.49\times$	$1.32\times$	$1.49\times$	$1.32\times$	$1.04\times$	$1.01\times$
LSM $h_z = 1.6$	B (heavy-hex)	$1.49\times$	$1.13\times$	$1.49\times$	$1.13\times$	$0.89\times$	$0.90\times$
<i>mean</i>	B (heavy-hex)	$1.87\times$	$1.62\times$	$1.81\times$	$1.51\times$	$1.24\times$	$1.27\times$
MFIM $h_z = 0$	C (chain)	$2.24\times$	$1.79\times$	$2.06\times$	$1.62\times$	$1.48\times$	$1.40\times$
MFIM $h_z = 0.5$	C (chain)	$2.11\times$	$1.66\times$	$1.98\times$	$1.73\times$	$1.49\times$	$1.34\times$
LSM $h_z = -0.7$	C (chain)	$1.00\times$	$1.00\times$	$1.00\times$	$1.00\times$	$0.71\times$	$0.55\times$
LSM $h_z = 1.6$	C (chain)	$1.00\times$	$1.00\times$	$1.00\times$	$1.00\times$	$0.54\times$	$0.55\times$
<i>mean</i>	C (chain)	$1.59\times$	$1.36\times$	$1.51\times$	$1.34\times$	$1.05\times$	$0.96\times$

$h_z = 1.6$ on Topology A, where our fidelity is lower), i.e. orders of magnitude inside the 1% tolerance; in 5 of the 12 cells our fidelity is in fact the higher of the two. The minimum fidelity attained anywhere along any CRAG-AVQDS trajectory is 99.92%. This is a structural consequence of the design rather than a lucky outcome: the operators we prefer are, by construction, operators that AVQDS itself certified as acceptable.

Topology A: the constraint is free, and slightly better than free. The square lattice admits a Hamiltonian cycle, so the a-priori placement puts every MFIM ring bond and every LSM chain bond on an edge and the naive baseline inserts *zero* SWAPs: its multiplier is exactly $1.00\times$. CRAG-AVQDS also inserts zero SWAPs, and reaches $0.66\times$. A multiplier below one is not a routing effect — there is nothing to route — but the single-qubit preference of the cost oracle (Sec. 2.3) operating inside

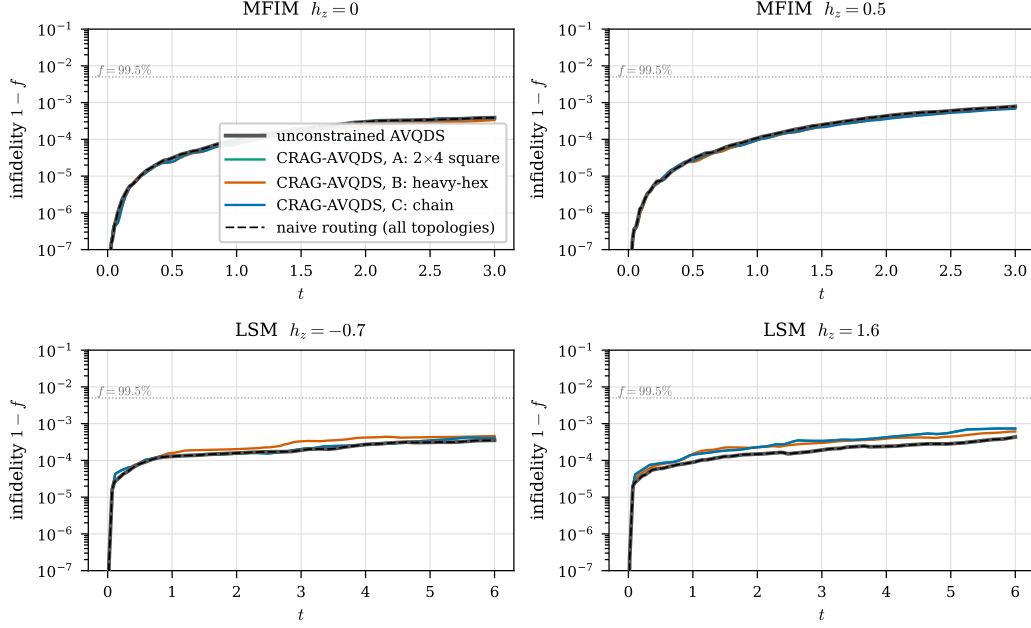


Figure 3: Infidelity $1 - f$ against time for the four benchmarks. The smart-embedding naive baseline routes the unconstrained circuit *post hoc*; SWAPs are exact unitaries and the residual placement is a classical relabelling, so its state — and hence its fidelity curve — coincides with unconstrained AVQDS to machine precision on every topology (dashed black, plotted over the grey curve). Only CRAG-AVQDS changes the state, because it changes which operators enter the ansatz.

the acceptance set: with every two-qubit pool operator costing the same 2 CNOTs, Eq. (6) reduces to “take an acceptable weight-one generator if one exists”. It is a genuine reduction of the logical gate count, but it is not a topology result, and we keep it out of the topology claim by isolating it in the ablations of Sec. 5.1. Topology A is therefore best read as a control: it confirms that a good placement plus a standard router is already optimal when the interaction graph embeds, and that our layer does no harm there.

Topology B: where the routing problem has bite. On the heavy-hex tree, post-hoc routing of the unconstrained circuit costs $1.87\times$ the CNOTs on average over the four benchmarks, and $1.81\times$ even when granted the oracle placement chosen by exhaustive search over all $8!$ bijections to minimise its own realised cost. CRAG-AVQDS costs $1.24\times$: a 34% reduction against the naive baseline and a 32% reduction against the oracle-placed one. The worst single (benchmark, topology B) multiplier for our method is $1.67\times$.

Topology C: the third graph. The linear chain is included because it isolates a variable that the first two graphs confound. Its total routing pressure for the MFIM ring, $\sum_b d_G = 14$, is *identical* to Topology B’s, but it is concentrated: seven bonds local and one at distance seven, rather than B’s five local, two at distance two and one at distance five. For LSM the chain is perfect, like Topology A. The multipliers are $1.59\times$ (naive), $1.51\times$ (oracle) and $1.05\times$ (ours), and the contrast with B is analysed in Sec. 5.1.

4.4 Physical observables

Figures 5 and 6 confirm that the constraint does not distort the physics. For MFIM the Loschmidt echo, which is sensitive to the full many-body wavefunction and is the observable used to identify dynamical quantum phase transitions Heyl et al. [2013], is reproduced by the topology-aware circuits at the same visual accuracy as by the unconstrained ansatz. For LSM the instantaneous energy tracks the exact ramp through the transition and into the post-ramp regime, and the crossover from xx - to yy -dominated correlations — including the fact that the long-range $\langle \sigma_0^y \sigma_7^y \rangle$ lags far behind its

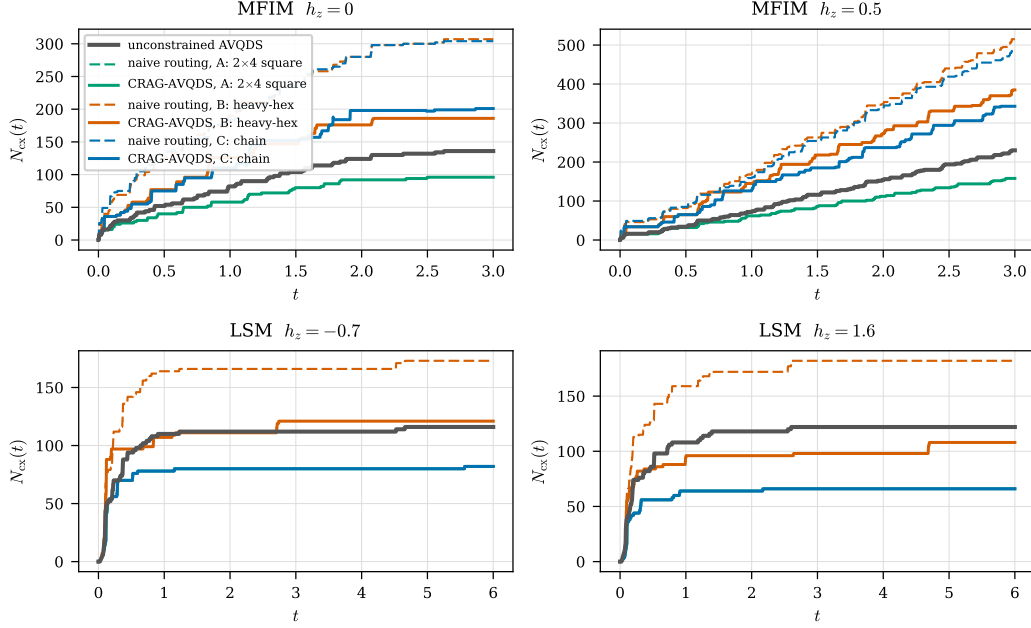


Figure 4: CNOT count of the compiled circuit that prepares $|\Psi(t)\rangle$, against t . Each point is a full re-compilation of the ansatz prefix at that time, so the curves are the circuits one would actually run, not extrapolations. Grey: unconstrained AVQDS. Dashed: smart-embedding naive routing. Solid coloured: CRAG-AVQDS. Colour encodes topology. On Topology A the interaction graph embeds exactly, so naive routing inserts no SWAPs and its curve lies exactly on the unconstrained one; on Topology C the same is true for LSM, whose open chain is a subgraph of the device chain.

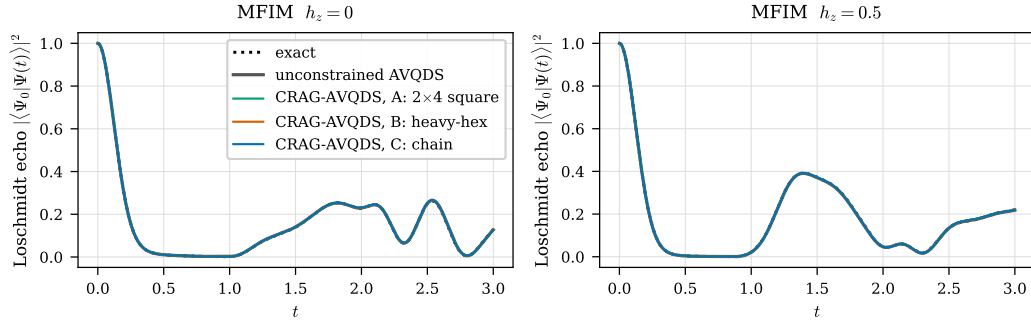


Figure 5: Loschmidt echo $|\langle\Psi_0|\Psi(t)\rangle|^2$ for the two MFIM quenches. The topology-aware circuits track the exact return probability, including the non-analytic dips associated with dynamical quantum phase transitions [Heyl et al. \[2013\]](#), [Heyl \[2018\]](#), as closely as the unconstrained ansatz.

equilibrium value because the ramp is non-adiabatic — is captured identically. This is the expected outcome given the fidelities, and we report it because an ansatz that matched a global fidelity while distorting local observables would be a real failure mode.

5 Analysis

5.1 Mechanism: what the acceptance slack buys

The slack is real and it is used. The diagnostics logged at every growth event show that in 85% of them the sufficient set $\mathcal{S}$ is non-empty — some single pool operator already brings L^2 below L_{cut}^2 — so in the large majority of decisions the accuracy criterion leaves the choice open and Eq. (6) is

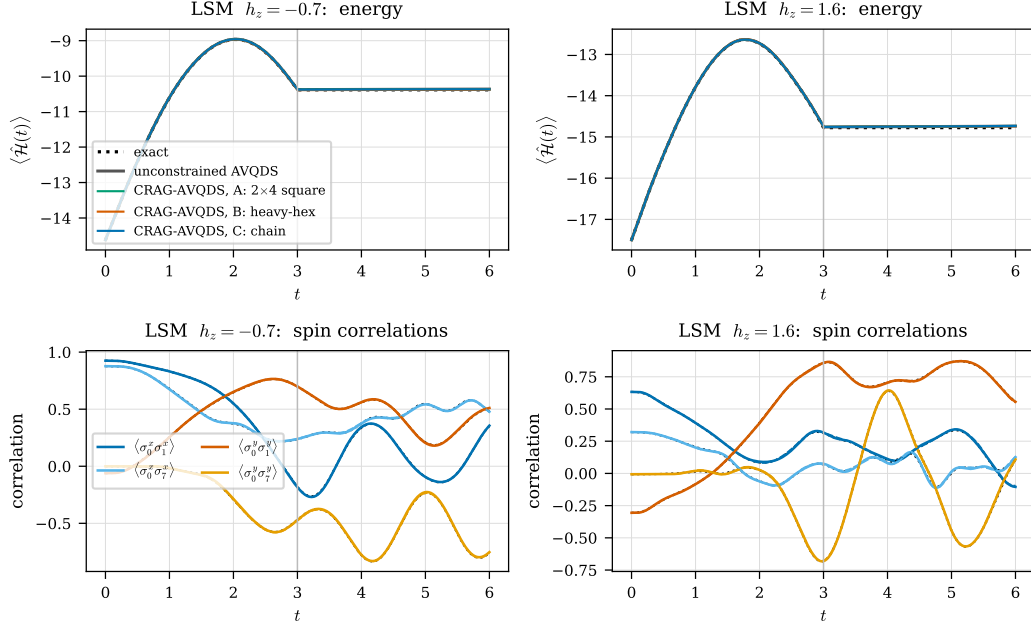


Figure 6: LSM benchmarks. Top: instantaneous energy through the ramp ($t \leq 3$, vertical line) and the post-ramp evolution. Bottom: the four spin correlators; solid, unconstrained AVQDS; dashed, CRAG-AVQDS on Topology B; dotted black, exact.

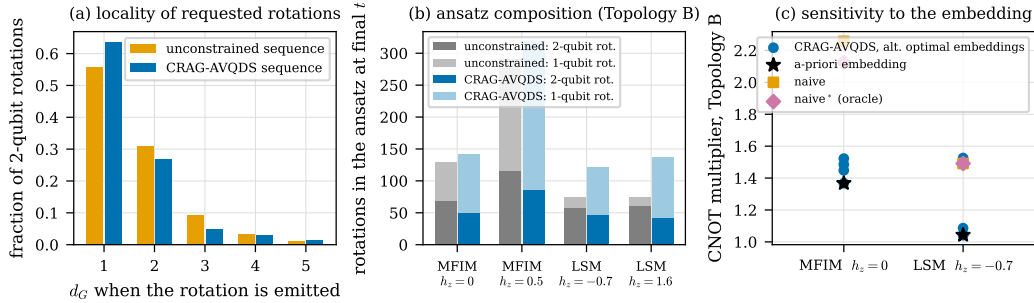


Figure 7: (a) Distribution of the graph distance between the two qubits a weight-2 rotation acts on, measured at the moment the rotation is emitted — i.e. under the placement the router has reached at that point — on Topology B, pooled over the four benchmarks. Orange: the operator sequence unconstrained AVQDS produces (the naive baseline must route this). Blue: the sequence CRAG-AVQDS produces. (b) Composition of the final ansatz: how many one- and two-qubit rotations each method ends with. (c) Sensitivity of the Topology-B CNOT multiplier to the placement, over placements drawn from the tie class of Eq. (8); the star marks the a-priori placement used everywhere else in the paper, and the naive baselines are shown for scale.

choosing among candidates that are, by AVQDS’s own standard, equally good. Only the remaining 15% of events fall through to the tolerance band. Neither safeguard of Sec. 2.4(b) matters in practice: the inner-cycle revert fired 0 times and the stall guard 0 times across all reported runs, so no result depends on either.

The gain is a shift in what the ansatz is made of. Figure 7(a) makes the mechanism directly visible. The quantity to measure is the graph distance between a rotation’s two qubits *at the moment it is emitted*, under the placement the router has actually reached — not under the fixed initial placement, which is the wrong reference for a method whose placement drifts as SWAPs accumulate, and drifts differently for the two methods. Under that measure, on Topology B the rotations unconstrained

AVQDS requests sit at mean distance 1.58 with 57% already nearest-neighbour, and the router has no choice but to buy the rest with SWAPs; the sequence CRAG-AVQDS produces sits at mean distance 1.44 with 68% already nearest-neighbour. The circuit is not routed better — it is the same router, with the same configuration search — it simply needs less routing. Figure 7(b) shows the second half of the story: the total number of two-qubit rotations itself falls to $0.75\times$ the unconstrained value, with the shortfall taken up by one-qubit rotations, which cost no CNOTs at all.

The two effects add up transparently, because our CNOT count decomposes exactly into $2n_{\text{rot}2} + 3n_{\text{swap}}$. On Topology B the logical part alone is $0.75\times$ the unconstrained count — *below* one, i.e. our circuit would be cheaper than the unconstrained one even with all-to-all connectivity — and routing adds a further $0.49\times$ (these are means of per-benchmark ratios, and they sum to the $1.24\times$ of Table 3 because the decomposition is exact benchmark by benchmark). Averaged the same way, SWAPs are 36% of the gates we emit. The naive baseline starts from a logical part fixed at $1.00\times$ by construction and can only add routing on top; that is the whole of the difference.

Decomposing the two effects. These are two distinct mechanisms and we separate them with two ablations (Fig. 2, right). (i) *Cost-blind to distance*: keep the acceptance rule but flatten c_σ to 2 for every weight-2 operator, so only the preference for weight-one generators survives. Over the 4 benchmarks we ran it on (MFIM $h_z = 0$, MFIM $h_z = 0.5$, LSM $h_z = -0.7$, LSM $h_z = 1.6$) this gives a Topology-B multiplier of $1.26\times$, against $1.87\times$ for naive routing, $1.81\times$ for the oracle-placed baseline and $1.24\times$ for the full method. (ii) *Cost-regularised growth at all-to-all connectivity*: run the same rule with no graph at all; it reaches $0.66\times$ the unconstrained CNOT count at a worst-case fidelity difference of 0.030 % — and on the square lattice it reproduces the Topology-A circuit gate for gate, confirming that the sub-unity multiplier there is entirely the weight-one preference and has nothing to do with connectivity.

Splitting our advantage over the oracle-placed baseline between the two mechanisms gives 98% from preferring weight-one generators and 2% from graph-distance awareness. We expected the second to dominate on a tree. It does not, and this is the single most important qualification on the paper. The weight-one preference is a real saving but a topology-*independent* one — an unconstrained implementation could take it too, and the “cost-regularised, all-to-all” row of Table 5 shows it doing exactly that — so the part of our margin that is genuinely about connectivity is much smaller than the headline multiplier suggests.

Worse for the distance term: on 2 of the 4 benchmarks the cost-blind variant is *better* than the full method (MFIM $h_z = 0.5$, $1.55\times$ against $1.67\times$; LSM $h_z = 1.6$, $0.84\times$ against $0.89\times$), so switching graph-distance awareness on costs CNOTs there. Averaged over the four the full method is still ahead ($1.24\times$ against $1.26\times$), but by 0.02 — a margin 2 of the 4 benchmarks individually reverse. A plausible cause is an interaction with the frozen η : MFIM $h_z = 0.5$ is exactly the benchmark on which $\eta = 0.20$ is a poor choice (Table 4), and the cost-blind rule, seeing a flat cost landscape, is insensitive to that. We have not isolated the cause and do not claim to have. We report this rather than quietly ablating at a per-benchmark η , and we would not claim the distance term is established as beneficial on a benchmark-by-benchmark basis — only in the mean, and only weakly.

Why the square lattice and the heavy-hex tree behave so differently. Topology A admits a Hamiltonian cycle, so the MFIM ring and the LSM path embed with every pool bond on an edge: the routing problem is empty, the naive baseline is already optimal at $1.00\times$, and any method that only re-routes must tie. This is the regime in which “topology-aware AVQDS” has nothing to do, and it is worth stating plainly that on such devices post-hoc routing with a good placement is the right engineering answer.

Topology B is a tree with four leaves. A Hamiltonian path on n vertices has at most two degree-one endpoints, so none exists; and for the periodic ring a stronger counting argument applies — the eight bonds form a closed walk, every tree edge must be crossed an even number of times, hence $\sum_b d_G \geq 2|E| = 14$, attained. Routing is therefore unavoidable *for any placement*, which is exactly why the margin is largest here. The key structural point is that the pool is over-complete for the task: the MFIM Hamiltonian has eight ZZ bonds but the growth loop, at any given event, does not need a particular one. Correlations that would naturally be generated across the expensive bond can be generated instead by a composition of rotations on cheap bonds, at the price of a slightly larger N_θ ($1.42\times$ on average) but a much smaller CNOT count. Group-theoretic intuition supports this: the Lie

algebra generated by the nearest-neighbour subset of the pool already closes on a large subalgebra, so restricting *preference* (not membership) costs expressivity slowly.

Topology C was included to isolate a variable, and it does. For the MFIM ring the chain carries *exactly the same* total routing pressure as the heavy-hex tree, $\sum_b d_G = 14$ under the optimal placement in both cases, but distributes it completely differently: C puts seven bonds at distance one and a single bond at distance seven, whereas B spreads the excess over three non-local bonds (distances 2, 2, 5). If total pressure were the controlling quantity, the two should behave alike. If *avoidability* is what matters, C should be the easier graph for us and not for the baseline, because one expensive generator out of eight can simply be used less often, whereas B’s three cannot all be dodged at once.

Restricting to the two MFIM benchmarks, where the pressures are equal by construction, our multiplier is $1.48\times$ on C against $1.52\times$ on B, while the oracle-placed baseline sits at $2.02\times$ and $2.14\times$. The ordering is in the direction the avoidability reading predicts for both methods, but the gap on our side is 0.04, and we measured a spread of $1.37\text{--}1.52\times$ on a single cell merely from changing the placement within its tie class. A difference four times smaller than the placement noise is not evidence, and we do not claim it as such: this experiment does not resolve whether concentration or total pressure is the controlling quantity. What Topology C *does* establish is the generalisation claim — the method transfers to a third graph with no retuning, giving $1.05\times$ against $1.51\times$ for the oracle-placed baseline — and it confirms the trivial-embedding control a second time, since LSM on the chain reproduces the Topology-A circuit exactly. Settling the concentration question would need either many more placements per cell or graphs whose pressure profiles differ far more sharply than these two.

Why this is not just better routing. Both sides of the comparison use the identical post-hoc router with the identical six-configuration search, and the baseline additionally gets an exhaustively optimised placement that our method does not. The margin therefore cannot be attributed to the routing stage; it is attributable to the operator sequence, which is what Fig. 7(a) measures directly.

5.2 Failure analysis

Depth does not track the CNOT count. Averaged over the four benchmarks our depth multipliers are $0.70\times$, $1.27\times$ and $0.96\times$ on the three topologies, against $1.00\times$, $1.62\times$ and $1.36\times$ for naive routing. Measured against the stronger oracle-placed baseline on the same cell, we reach 0.67 of its CNOT count but 0.74 of its depth — a proportionally smaller gain in depth than in gate count, with our depth worse than that baseline’s in 0 of 12 cells. The gap is systematic rather than incidental. The cost oracle of Eq. (4) prices CNOTs, and its cheapest single move is to substitute a one-qubit rotation for a two-qubit one. A one-qubit rotation occupies a layer on its own qubit and cannot be parallelised against anything else on that qubit, whereas the two-qubit rotation it replaced could often have run concurrently with gates on a disjoint pair. The substitution is therefore strictly good for N_{ex} and only weakly good — occasionally neutral — for the critical path. A depth-aware oracle, pricing a candidate by the increase in the ASAP frontier over its support rather than by CNOTs, is the natural fix and is a one-function change in our framework; we did not pursue it because the CNOT count is the accepted proxy for near-term feasibility, and we would rather report the mismatch than retune the objective after seeing the results.

One global hyperparameter costs something on the benchmarks it was not tuned on. η was selected on MFIM $h_z = 0$ and then frozen. Running the full grid on the other benchmarks (Appendix Table 4) shows what that costs: over the 4 configurations with a complete grid (MFIM $h_z = 0$, MFIM $h_z = 0.5$, LSM $h_z = -0.7$, LSM $h_z = 1.6$), the frozen value gives a mean Topology-B multiplier of $1.24\times$ where per-benchmark tuning would have given $1.12\times$. The largest single gap is on MFIM $h_z = 0.5$, where $\eta = 0.50$ reaches $1.45\times$ against the $1.67\times$ of the frozen value. We report the frozen value in every table, because a per-benchmark choice would be tuned on the test set; but the gap is a real headroom that a practitioner with a calibration budget could recover, and it also shows the response to η is not monotone — the band interacts with the discrete growth decisions in a way our current analysis does not predict.

A numerical-library caveat, not a method limitation. One grid point of the η sweep ($\eta = 0.5$ on LSM $h_z = 1.6$, Topology B) initially terminated inside the trial-expansion loop with a non-convergent singular-value decomposition in the Tikhonov-regularised pseudo-inverse $(M + \xi\mathbb{I})^{-1}$. Repeating

the identical, deterministic run with a different BLAS thread count completed it normally, so the cause was LAPACK’s convergence behaviour on this matrix and not a property of the trajectory: the figure is in Table 4 like every other. We record it because it is a reproducibility hazard worth knowing about — the McLachlan matrix at the widest band we tried is close enough to singular that an SVD implementation detail decides whether the run finishes — and because we would rather report a transient failure and its resolution than quietly rerun until it passed. It does not occur at the selected $\eta = 0.20$, and no headline run was affected.

The method needs slack, so it does nothing when there is none. At early times N_θ is small, L^2 is far above L_{cut}^2 , and the growth loop must add several operators per event; the sufficient set is empty and only the tolerance band applies. This is visible in Fig. 4 as the initial coincidence of the curves. Conversely a device on which every pool operator costs the same — Topology A — leaves the distance term inert. The method’s benefit is therefore bounded above by the size of the acceptance set, which is a property of the model and the pool, not something we control.

N_θ inflation is a real cost in the measurement budget. Our circuits are cheaper in gates but carry more parameters ($1.42\times$ on average on Topology B, up to $1.85\times$). For AVQDS on hardware the number of distinct measurement circuits scales as $\mathcal{O}(N_{\mathcal{H}}N_\theta + N_\theta^2)$ Yao et al. [2021], so a $1.42\times$ increase in N_θ is roughly a 1.42^2 -fold increase in the tomographic cost of one time step. Whether this is a good trade depends on the device: on a gate-error-limited processor, removing SWAPs is worth a great deal and extra measurements are cheap; on a shot-limited one, the balance can invert. We regard this as the most important caveat on the headline numbers, and it is not visible in any gate-count table.

Placement dependence. Figure 7(c) shows the spread of the Topology-B CNOT multiplier over placements drawn from the tie class of Eq. (8), on 2 benchmarks (MFIM $h_z = 0$, LSM $h_z = -0.7$). Pooled, the multiplier ranges over 1.04 – $1.53\times$ with mean $1.32\times$; on MFIM $h_z = 0$ alone the range is 1.37 – $1.52\times$ against the $1.37\times$ of the a-priori placement used in the tables, and on LSM $h_z = -0.7$ it is 1.04 – $1.53\times$ against $1.04\times$.

This is the largest single source of uncertainty in the paper, and it cuts against us. The a-priori placement is the best of its tie class on MFIM $h_z = 0$ and the best on LSM $h_z = -0.7$ — on both benchmarks sampled it is the most favourable member. Our headline numbers therefore sit at the favourable end of their own tie classes, and a reader should compare the baseline against the *range* rather than the single reported value. More sharply: on 1 of the 2 benchmarks swept, the *worst* placement in the tie class (LSM $h_z = -0.7$: $1.53\times$) is beaten by the oracle-placed naive baseline itself ($1.49\times$). A practitioner who applied our rule and drew an unlucky member of the tie class could therefore end up behind a well-placed post-hoc router on that benchmark. The a-priori rule of Eq. (8) is under-determined — it leaves a large tie class and does not discriminate within it — and making it discriminate is, on this evidence, at least as valuable as anything else we could add to the method.

Cutting the other way, the asymmetry in placement search is against us: the baseline is granted an exhaustive search over all $8!$ placements while we take whatever the a-priori rule returns. We did not sample placements on the two remaining benchmarks, and we would not defend any claim resting on differences smaller than this spread — which is exactly why we decline to read the Topology B/C comparison above as evidence.

Where the margin is smallest, and why the average flatters us. Four of the 12 (benchmark, topology) cells are not really contests. On Topology A the interaction graph embeds, the naive baseline inserts no SWAPs and sits exactly at $1.00\times$, and any difference we show there comes from the topology-independent single-qubit preference rather than from routing. The same holds for LSM on Topology C, whose open chain is a subgraph of the device chain. Averaging the multiplier over all twelve cells therefore mixes cells where the routing problem is empty with cells where it is not, and it is the Topology-B column — the only one where a routing problem exists on every benchmark — that carries the claim. The honest summary is that this method matters on sparse, tree-like connectivity, which is precisely the heavy-hex regime of current superconducting hardware, and is a modest topology-independent CNOT saving elsewhere.

5.3 Computational cost

The topology layer’s own cost is one table lookup per pool candidate per growth event: $|\mathcal{P}| \leq 24$ lookups against a precomputed 8×8 distance matrix, negligible beside the $|\mathcal{P}|$ bordered pseudo-inversions that the growth event performs anyway (each $\mathcal{O}(N_\theta^3)$). Instrumenting the selection routine directly, it accounts for 0.001% of wall time.

To measure the ratio rather than the absolute cost, each benchmark’s pair (unconstrained first, then CRAG-AVQDS on Topology B) is executed back to back *inside a single process* with a fixed thread count, so that both members see the same machine load; the machine is shared with the rest of the batch, so absolute wall times are contended and only the ratio is meaningful. Over the 4 benchmarks measured this way (LSM $h_z = -0.7$, LSM $h_z = 1.6$, MFIM $h_z = 0$, MFIM $h_z = 0.5$), the total wall-clock ratio is 1.25 on average, ranging over 0.81–1.91; the per-step ratio is 1.32. Growth events account for 1.4% of wall time, the rest being the EOM assembly that both methods share. The uncontrolled batch runs give a consistent picture (0.98 on average across all twelve topology-aware runs, spread 0.54–1.31).

Two effects push in opposite directions, which is why the ratio straddles unity rather than sitting cleanly above it, though they do not cancel: the mean is 1.25, not 1. Our ansatz carries more parameters ($1.42\times$ on Topology B), and the dominant cost of an AVQDS step — building the N_θ derivative states — scales as N_θ^2 , so a larger N_θ is more expensive per step; this is the effect that dominates. Against that, the runs take a different number of steps (0.96 on average, up to 1.10), because the adaptive step size $\delta t = \delta\theta_{\max}/\|\dot{\theta}\|_\infty$ responds to the conditioning of the ansatz, and our tie-break retains the reference implementation’s preference for well-conditioned candidates; on one benchmark this makes CRAG-AVQDS the *faster* of the two. We would not read anything into the sign of the difference — it is a second-order effect of the step controller, not a property of the selection rule — only into its magnitude, which is small.

The practical conclusion is that CRAG-AVQDS sits within a small constant factor of unconstrained AVQDS on the classical cost–quality curve — not at the same point, but close enough that classical time is not the binding constraint — while producing hardware-executable circuits. This matters for the comparison with the alternative strategy of solving the placement problem harder: the oracle placement we granted the naive baseline required $8! = 40320$ full routings per (benchmark, topology), which is affordable at $N = 8$ and combinatorially hopeless beyond it, whereas the cost of our rule is linear in the pool size and independent of N except through the distance matrix.

6 Discussion and conclusion

Adaptive variational algorithms occupy an unusual position with respect to compilation. In every other setting the compiler receives a circuit and must transport it onto a sparse device; here the circuit does not exist yet when the device’s constraints could most cheaply be honoured. We have shown that the decision point at which an adaptive dynamics algorithm chooses its next generator carries enough freedom to absorb most of the connectivity penalty, and that using it beats routing the unconstrained circuit even when the routing stage is given the best placement obtainable by exhaustive search.

Three things make the freedom usable rather than merely present. First, the growth loop’s stopping criterion is an inequality, so in 85% of growth events more than one operator discharges the accuracy obligation and the algorithm is already tie-breaking on a secondary heuristic — we simply change what the heuristic is. Second, the McLachlan improvements $\{\Delta_\nu\}$ across a Hamiltonian pool are frequently near degenerate, a fact reported in the original AVQDS work as an obstacle (it makes enlarged pools produce longer circuits) and used here as a resource. Third, because a substituted operator is one that AVQDS itself certified, the accuracy cost is bounded by construction: across 12 (benchmark, topology) cells the largest final-time fidelity difference from the unconstrained baseline is 0.030 % absolute.

The measured outcome is that the CNOT cost of connectivity on an eight-qubit heavy-hex tree drops from $1.87\times$ (post-hoc routing, a-priori placement) or $1.81\times$ (post-hoc routing, oracle placement) to $1.24\times$, with fidelity, Loschmidt echo, energy and spin correlations indistinguishable from the unconstrained ansatz. The selection rule costs 0.001% of wall time; the larger ansatz it builds brings the total classical cost to $1.25\times$. On a 2×4 square lattice, where the interaction graph embeds exactly, there is nothing to route and the honest conclusion is that a good placement plus a standard

router is already optimal; our method still reduces the logical CNOT count slightly, but for a reason (a preference for single-qubit generators) that has nothing to do with topology and that we isolate rather than fold into the headline.

Limitations. Four deserve emphasis. (i) Ablation attributes only 2% of our advantage over the oracle-placed baseline to graph-distance awareness, the rest coming from a preference for weight-one generators that is real but topology-independent; on 2 of the 4 ablated benchmarks the distance term is net negative. The headline multiplier is therefore not all attributable to the topology-aware idea, and we say so in Sec. 5.1 rather than in a footnote. (ii) The gain in *depth* is smaller than the gain in gate count, because our cost oracle prices CNOTs and its cheapest move — substituting a one-qubit generator — can lengthen the critical path. (iii) The circuits carry more variational parameters ($1.42 \times$ on Topology B), and since the AVQDS measurement budget grows as $\mathcal{O}(N_\theta^2)$ per step, the method trades gate error against sampling cost in a way that is favourable on a gate-error-limited device and can be unfavourable on a shot-limited one. (iv) All results are at $N = 8$ with exact statevector simulation and no noise model, so nothing here speaks to how the method interacts with the error rates and coherence budgets of a real superconducting device Bravyi et al. [2022]; nothing here establishes how the margin scales, and the tree-versus-cycle structural argument, while size-independent in principle, has only been tested on three graphs.

Outlook. The framework isolates the contribution in a single function, the cost oracle c_σ , and this is where the natural extensions live. Replacing CNOT count by the ASAP-frontier increase gives a depth-aware variant; replacing it by a calibrated two-qubit error rate gives a noise-adaptive one in the spirit of Ref. Murali et al. [2019], which on real hardware may matter more than either; combining it with circuit-compression techniques that trade classical optimisation for depth Berthussen et al. [2022], Benedetti et al. [2021] is a further natural step. A larger question is whether the same acceptance slack can be spent on a different resource entirely — measurement cost, for instance, by preferring generators whose M -matrix elements are already being estimated — which would attack the N_θ^2 tomographic bottleneck that our own results identify as the dominant remaining cost. More broadly, the observation that an adaptive variational algorithm’s accuracy criterion under-determines its circuit, and that the residual freedom is a resource for hardware constraints, is not specific to McLachlan dynamics; the same structure appears in ADAPT-VQE, in adaptive imaginary-time evolution Gomes et al. [2021], and in projected variational dynamics Barison et al. [2021], and we expect it to be exploitable there for the same reason.

Reproducibility. Every run reported here is in `proposal/logs/runs/` as a JSON record with hyperparameters, per-step time series, the full operator sequence, the compiled circuit’s per-pair two-qubit audit and the statevector cross-check; the tables and all numbers quoted in the text are generated from those records by `proposal/code/aggregate.py` and `macros.py`. The method is deterministic — no random number generator is used by AVQDS, by the selection rule, by the router or by the placement search — so runs are bit-reproducible.

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A Supporting tables

Nothing was relegated to this appendix because it fell short of the standard of the main text. The material below is the complete supporting evidence for claims made in Sections 4 and 5: the full hyperparameter sweep with no run withheld, the component ablation, the placement-sensitivity spread, and the aggregate compliance audit. Every entry is generated from `proposal/logs/runs/` by `proposal/code/aggregate.py`.

Table 4: Complete η sweep on Topology B: every value tried on every benchmark listed, with no run withheld. N_{cx} multiplier against unconstrained AVQDS, and the absolute final-time fidelity difference from it. η was selected on MFIM $h_z=0$ alone and then frozen; the remaining columns are an out-of-sample sensitivity check, and show both that the accuracy cost of the band is negligible everywhere and that the response to η is not monotone. $\eta=0$ retains the ‘cheapest sufficient operator’ branch and is therefore already topology-aware.

η	MFIM $h_z=0$		MFIM $h_z=0.5$		LSM $h_z=-0.7$		LSM $h_z=1.6$	
	N_{cx} mult.	Δf	N_{cx} mult.	Δf	N_{cx} mult.	Δf	N_{cx} mult.	Δf
0.00	1.52×	5.8×10^{-5}	1.55×	6.6×10^{-5}	0.89×	1.8×10^{-4}	0.80×	1.1×10^{-4}
0.05	1.40×	7.0×10^{-5}	1.67×	4.2×10^{-5}	0.91×	3.3×10^{-4}	0.86×	5.5×10^{-4}
0.10	1.40×	7.0×10^{-5}	1.67×	4.2×10^{-5}	0.87×	5.8×10^{-6}	0.86×	2.9×10^{-4}
0.20*	1.37×	5.3×10^{-5}	1.67×	4.2×10^{-5}	1.04×	1.1×10^{-4}	0.89×	1.8×10^{-4}
0.30	1.37×	5.3×10^{-5}	1.46×	6.1×10^{-5}	0.97×	8.2×10^{-5}	1.01×	2.7×10^{-4}
0.50	1.52×	8.5×10^{-5}	1.45×	7.9×10^{-5}	1.04×	1.4×10^{-4}	0.89×	3.7×10^{-4}

*selected by the rule of Sec. 2.7.

Table 5: Component ablation. ‘cost-blind to distance’ keeps the acceptance rule of Eq. (6) but flattens the oracle to $c_\sigma=2$ for every weight-2 rotation; ‘cost-regularised, all-to-all’ applies the rule with no connectivity graph at all and is compiled under full connectivity. Values are N_{cx} multipliers against unconstrained AVQDS (Topology B for the first four rows).

Configuration	MFIM $h_z=0$	MFIM $h_z=0.5$	LSM $h_z=-0.7$	LSM $h_z=1.6$	mean
Smart-embedding naive routing (B)	2.26×	2.24×	1.49×	1.49×	1.87×
+ oracle placement (B)	2.12×	2.15×	1.49×	1.49×	1.81×
Cost-blind to distance (B)	1.52×	1.55×	1.15×	0.84×	1.26×
CRAG-AVQDS (B)	1.37×	1.67×	1.04×	0.89×	1.24×
Cost-regularised, all-to-all	0.71×	0.69×	0.71×	0.54×	0.66×

Table 6: Sensitivity of the Topology-B CNOT multiplier to the placement, over placements drawn from the tie class of Eq. (8). ‘a-priori’ is the placement actually used in the main tables; the rule does not select for being the best of its tie class, and the spread bounds how much of any reported margin could be placement luck.

Benchmark	#placements	min	mean	max	a-priori
LSM $h_z=-0.7$	4	1.04×	1.17×	1.53×	1.04×
MFIM $h_z=0$	4	1.37×	1.46×	1.52×	1.37×

Table 7: Topology-compliance audit, aggregated over every constrained circuit produced in this work (naive, oracle-placed naive, CRAG-AVQDS, all ablations and all sweep points). ‘ $\max|\Delta\psi|$ ’ is the largest amplitude deviation between the compiled circuit executed on an independent statevector simulator and the state the AVQDS core produced. Per-gate records are in `logs/compliance_audit_{A,B,C}.json`.

Topology	circuits	2-qubit ops	SWAPs	edge-set violations	$\max \Delta\psi $
A (square)	12	805	0	0	6.8×10^{-15}
B (heavy-hex)	44	3895	1342	0	6.8×10^{-15}
C (chain)	12	1168	356	0	6.8×10^{-15}

B Reproducing the results

The pipeline is six commands from `proposal/code/`:

1. `python tests/test_memoisation.py` — verifies that the performance patch and the topology subclass with the layer disabled reproduce the reference AVQDS trajectory bit-for-bit (identical step count, identical operator sequence, $\max|\Delta\theta| = \max|\Delta\Psi| = 0$).
2. `python tests/test_router.py` — the randomised compiler and router stress test of Sec. 3.5.
3. `python orchestrate.py -stage 1` — the four unconstrained baselines and the η grid on Topology B.
4. `python pick_eta.py` — applies the pre-registered selection rule and writes `logs/eta_star.json`.
5. `python orchestrate.py -stage prio then -stage extras` — CRAG-AVQDS on the remaining topologies, then the component ablations and the placement sweep.
6. `bash finalize.sh` — rebuilds `logs/summary.json`, the three main tables, the appendix tables, every macro used in the prose, all figures, the per-topology compliance audits and the side-by-side naive-routing audit, compiles `report.pdf`, and runs `check_paper.py`, which fails the build on an undefined macro, an unfilled marker token, an empty table cell, an unresolved citation, a hyperparameter mismatch between the paper and the logs, or a LaTeX error.

`run_rest.sh` chains steps 3–6 together with the controlled timing measurement. No step uses a random number generator, so the pipeline is bit-reproducible.

Each record in `logs/runs/` contains the job specification, the complete hyperparameter set actually used, the wall-clock time, the step count, the final-time observables, the full ordered operator sequence of the ansatz, and — for every compiled circuit derived from that trajectory — the gate counts, the depth, the placement, the final placement after routing, the per-pair two-qubit audit against the edge set, the router configuration selected, and the statevector cross-check residual. The companion `*_series.npz` holds the per-step time series (time, fidelity, N_θ , gate counts, McLachlan distance, energy, Loschmidt echo and, for LSM, the four spin correlators) for both the variational and the exact state.